LEVELHULL · RISK REGISTER AND HAZARD ANALYSIS

LevelHull risk register and hazard analysis

Design Molecule Lab · designmolecule.com

DOCUMENT CONTROL

Document ID	LVH-RSK-001	Version	0.2
Status	Draft	Date	2026-10-25
Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/levelhull		
Source	docs/09-risk.md @ e0fd582		

REVISION HISTORY

Version	Date	Author	Change
0.2	2026-10-25	Amish Chadha	H2 and H6 controls as decided by Amish (D-O2, D-O3), 2026-10-25
0.1	2026-10-24	Amish Chadha	First issue (kit 1.9.1 upgrade) with project risks, preliminary hazard analysis and a first-pass design FMEA

LevelHull risk register and hazard analysis

The top risks are that crews treat the kit as a reason not to wear life jackets (risk compensation), that covers and straps degrade in the sun until a module tears free when it is needed, and that the canoe is nearly awash in the first moments after swamping (about 24 mm freeboard until the water evens out). The design FMEA's highest RPNs are covers degrading unseen under the straps (RPN 75) and strap webbing losing strength in the sun (RPN 60); both are controlled by inspection, not by calculation. Method: STANDARDS section 23. Likelihood, impact, severity, occurrence and detection are scored 1 (low) to 5 (high; for detection, 5 is hardest to detect).

> **Safety:** LevelHull is flotation for the boat, never a substitute for life jackets. Every hazard below assumes life jackets are worn; the residual ratings do not hold if they are not.

Project risk register

Table 1. Project risks.

#	Risk	Cause	Effect	Likelihood	Impact	Owner (role)	Response
P1	No co-design partner	No BMU or institute takes part	No real hulls, loads or swamp test; stays at TRL 3	3	4	Project lead	Approach the three candidates in LVH-PRB-001 in order
P2	Owners reject the lost space	8.4 % of the inside volume high on the sides	Kits removed or never fitted	3	4	Design lead	Show option B to Amish (LVH-OPT-001); measure acceptance in the pilot
P3	Approval needed and slow	Kenyan rules on modifications (LVH-REG-001 G1)	Delay or redesign	3	3	Partner liaison	Ask the Kenya Maritime Authority early; offer the swamp test as evidence
P4	Foam not available at the assumed grade or price	Local supply of closed-cell PE sheet	Cost or lift changes	3	3	Design lead	Data sheet and samples before buying; calculator re-run for any density
P5	Cost over the value-engineering target	Foam is 48 % of the kit	USD 360 over for two canoes	4	2	Design lead	Savings in LVH-DEC-001 (bulk foam, used tarpaulin, galvanised in fresh water)
P6	Misleading promotion	Kit described as lifesaving	Fewer life jackets worn; liability	2	5	Project lead	PR-015: life jacket message on every document and kit

Preliminary hazard analysis

Table 2. Hazards.

#	Hazard	Cause	Harm	Severity	Likelihood	Control	Residual
H1	Risk compensation	Crews feel safe without life jackets	Drowning if thrown clear of the canoe	5	3	Message in every document and on the kit; promotion only with life jacket programmes (PR-015, UR-015)	Medium
H2	Canoe nearly awash just after swamping	Water inside trapped above the lake level (LVH-STM-001 S2)	Crew panic, further swamping in waves	4	3	Operating rule "hold on, then bail at once, seated" (LVH-DEC-001 D-O2, decided 2026-10-25); four bailers, about 11 min from brim-full (D-O3); 92 mm once the water evens out; freeing ports studied at TRL 4	Medium
H3	Module tears free	Strap, D-ring or cover failure; screws pull out	Loss of lift on one side, heel	5	2	Fixings at 5 or more times their share (S11, S12); pull test and proof load before use; annual inspection	Low
H4	Capsize in breaking waves	Waves larger than any passive flotation can resist	Crew in the water	5	2	Kit buys time, not safety; weather warnings; life jackets	Medium
H5	All crew on one side	Crowding at one grab line	Low-side gunwale 44 mm from the water	3	3	Grab lines on both sides; training to spread out	Low
H6	Instability while bailing	Free surface of the water inside (S18, 0.14 m metacentric height)	Re-swamping, capsize	4	2	Bail seated and low, two each side with the four bailers (D-O3)	Low
H7	Swamp test accident	Testing a filled canoe	Drowning, cold	5	1	Safety stops S1 to S5 in LVH-BLD-001: shallow sheltered water, safety boat, life jackets	Low
H8	Workshop injury	Saws, drills, solvents	Cuts, eye injury, fumes	3	2	Eye protection, ventilation (LVH-BLD-001)	Low
H9	Hull damage when fitting	Screw through the planking	Leak below the waterline	3	2	Screws stop 12 mm short of the planks; depth flag on the drill (S14, PR-017)	Low

Design FMEA (first pass, P3)

Table 3. Failure modes. RPN = severity x occurrence x detection.

#	Item	Function	Failure mode	Effect	Cause	S	O	D	RPN	Action
F1	Foam strips	Give lift	Water uptake, crushing	Less freeboard	Wrong foam grade; open cells; heat	4	2	3	24	Supplier data sheet; immersion samples (PR-006); 5 % uptake checked in CAL [F13]
F2	Foam strips	Give lift	Dissolved or swollen by fuel	Less lift	Polystyrene bought by mistake; fuel spill	4	2	2	16	Specify polyethylene, never polystyrene (BOM line 5); covers keep fuel off
F3	Covers	Shield the foam, hold the strips	UV breakdown, tearing	Foam exposed, strips lost	Tropical sun; abrasion by nets	5	3	2	30	Annual inspection; replace covers (tailor-made)
F4	Covers	Shield the foam	Covers degrade unnoticed under the straps	Module splits when loaded	UV and wear hidden at the strap	5	3	5	75	Inspection at each refit with the straps opened; spare covers; S10 open
F5	Straps	Hold the module down	Webbing loses strength	Module floats free when swamped	UV degradation of polyester	5	3	4	60	Replace straps on a schedule set from field data (PR-009); hand-pull check
F6	D-ring buckles	Close the strap	Slip	Strap loosens	Tail not tucked; worn webbing	4	2	3	24	Strap hold check (LVH-BLD-001 section 5)
F7	Coach screws	Carry the batten	Pull-out or corrosion	Batten and module free	Rot in the frame; weak frame timber; galvanised screws in tannin-rich hardwood	5	2	4	40	A4 stainless in salt water (S22); sealed holes; pull test; check frame fastenings (LVH-DEC-001 to confirm item 4)
F8	End chocks	Stop fore and aft creep	Chock splits or loosens	Module creeps against a thwart	Screws too short (fixed by D-S1)	2	2	2	8	Counterbored screws 31 mm into the batten (D-S1)
F9	Battens	Carry the module	Rot	Screw pull-out	Wet timber, unsealed ends	4	2	4	32	Durable hardwood, sealed ends and holes (BOM line 14)
F10	Grab lines and eye bolts	Something to hold	Line chafes through	Crew lose their hold	Abrasion on the strake	4	2	2	16	Inspect; replace rope

#	Item	Function	Failure mode	Effect	Cause	S	O	D	RPN	Action
F11	Whole kit	Float level	Fitted at the wrong height	Less roll stiffness or interference with thwarts	Gauge not made for that hull	3	2	2	12	Setting gauge per hull shape; fit check (PR-005)

F4 and F5 lead because degradation under a strap is nearly invisible without opening it; the response is an inspection routine, set from field data at TRL 4 and later.